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XRGait: Immersive Gait Training Visualization with Integrated Sensing

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Gait Training

Gait is the *pattern of movement*

Gait training focuses on

- Restoring normal walking patterns
- Improving strength, balance, and coordination
- Re-educating muscles



Problem Statement

- Traditional gait rehabilitation lacks real-time, accurate feedback and engaging user experiences.
(Bartloff et al., 2025)
- Most VR-based gait systems are designed for controlled clinical environments, not home-based training
(Lee et al., 2020, Cano Heinrich et al., 2022, Porras et al., 2018, Papegaaij et al., 2017, Lee et al., 2014)
- No direct modality comparison engagement, usability, and long-term adherence are often under-evaluated
(Stoutz et al., 2019, Caprani et al., 2015)

XRGait: Immersive Gait Training Visualization with Integrated Sensing



smart insoles for pressure and contact analysis,



wearable motion sensors for full lower-body tracking, and



immersive displays

to deliver real-time gait feedback inside an engaging training environment.

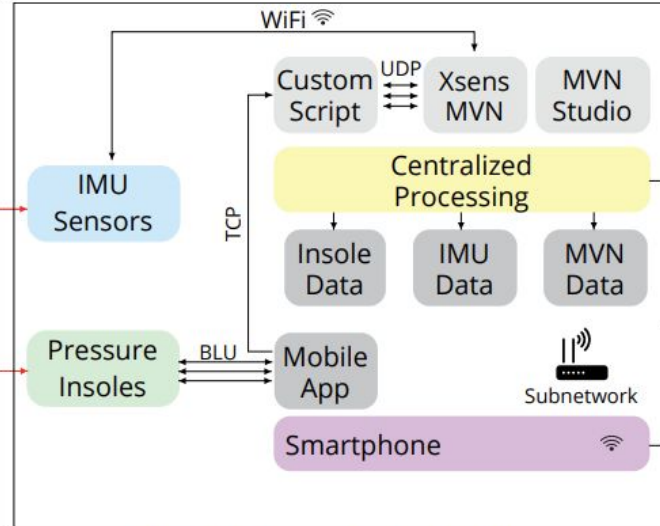


System Overview

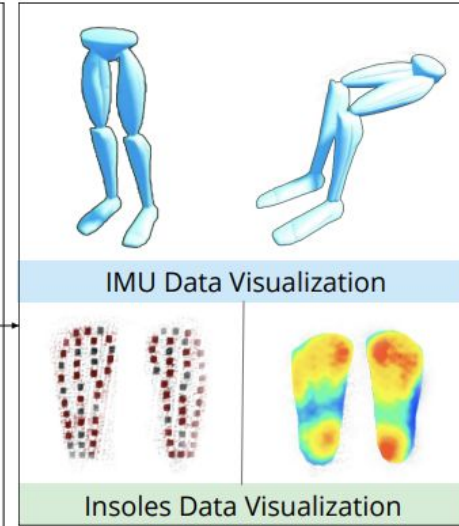
Smart Insoles



(a) Sensors alignment



(b) Overview of the data flow



(c) Visualisation and feedback



Training Environment

Applications

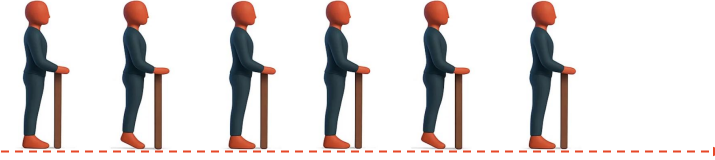


Exercises

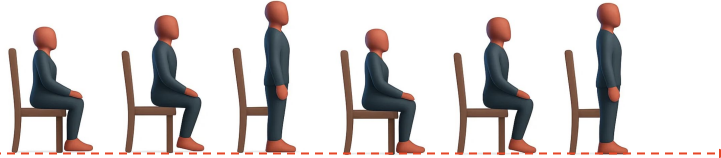
Seated Marching



Seated/Standing
Calf Raise



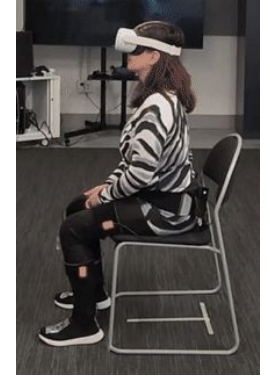
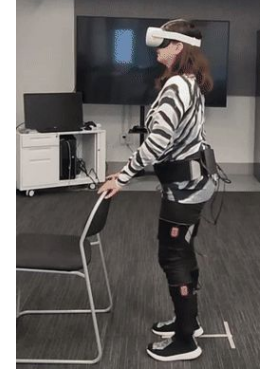
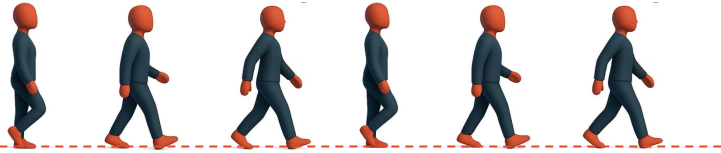
Sit to Stand



Standing
Flamingo



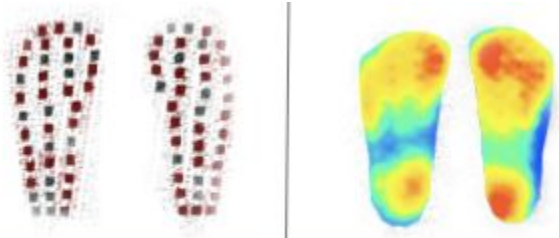
Walking



Study 1: Insoles + AR vs VR

Study 1:

- SensorDot vs HeatMap
 - AR vs VR
 - Within-subjects design
 - 14 participants
(Age: 21 to 34, $M = 26.4$, $SD = 4.6$)
- H1.** HeatMap visualisation will be rated significantly higher for usability than the SensorDot visualisation, and will be significantly preferred by users.
- H2.** The mobile AR will be rated significantly higher than the VR in both usability and user preference metrics



The image shows two foot-shaped visualizations of insole data. On the left, the SensorDot visualization consists of a grid of small red and black squares. On the right, the HeatMap visualization shows a color gradient from blue (low) to yellow (high) across the foot area.

Insoles Data Visualization		
	HeatMap	SensorDot
VR	VR with HeatMap Visualisation	VR with SensorDot Visualisation
AR	AR with HeatMap Visualisation	AR with SensorDot Visualisation

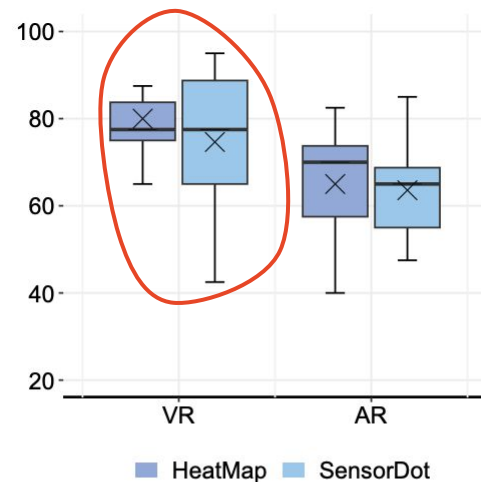
Study 1: Insoles + AR vs VR

Study 1:

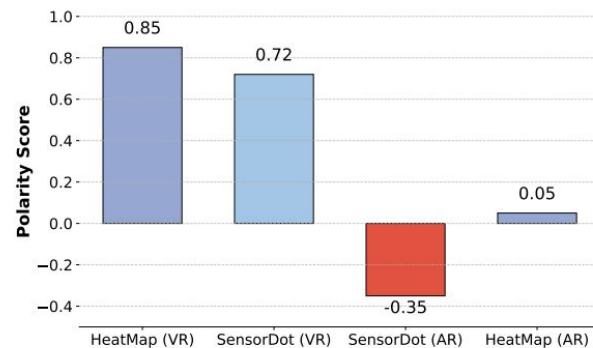
- ✓ VR more usable & engaging
- ✓ HeatMap clearer than SensorDot
- ✓ AR cumbersome for gait tasks (phone holding)
- ✓ Clinicians request lower-body tracking → informs iteration 2

“The HeatMap was easier for me to be like, oh, that’s what’s happening, like when I put pressure on my heel” — (P1; Study 1)

“The dots were not as clear, and I didn’t understand them as quickly” — (P6; Study 1)



(a) SUS by display type and visualisation.



(b) Sentiment Analysis in VR and mobile AR

Study 2: IMU + VR vs non-VR

Study 2:

- Lower-body rig
- VR vs Desktop
- 20 participants

(Age: 20 to 66, $M = 28.6$, $SD = 9.34$).

- H3.** The VR environment will lead to higher participant engagement, embodiment, and efficacy levels compared to the nonVR environment.
- H4.** The VR environment will involve increased task workload compared to the non-VR.



Study 2: IMU + VR vs non-VR

Study 2:

- ✓ VR significantly higher presence
- ✓ No TLX workload difference
- ✓ Visualizations useful especially for beginners
- ✓ Requests for more feedback, gamification

“The visualizer helped me understand how my body was moving and where I needed to make corrections” — (P9; Study 2)

“I liked being able to see my movements; it made it easier to correct my form, especially since I haven’t done much physiotherapy before” — (P20; Study 2)

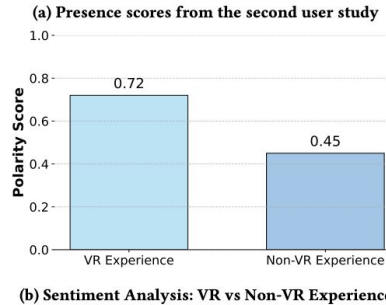
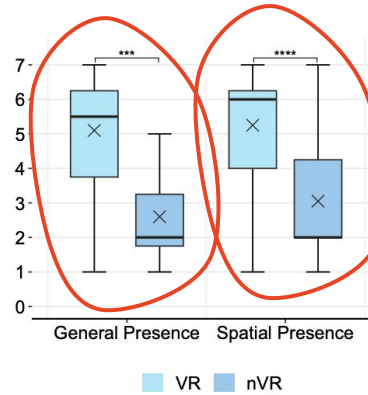


Figure 6: Study 2 results: (a) Presence and (b) sentiment analysis results across conditions.

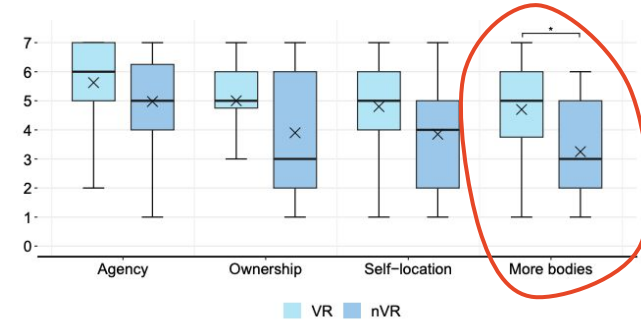


Figure 7: Embodiment questionnaire results from Study 2

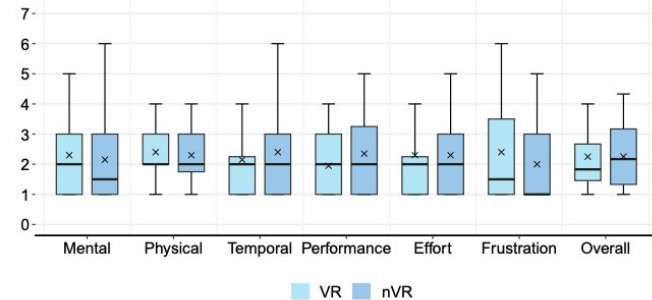
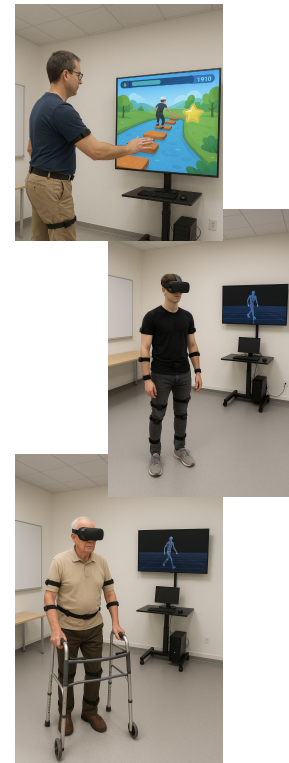


Figure 8: TLX results from Study 2 (lower is better)

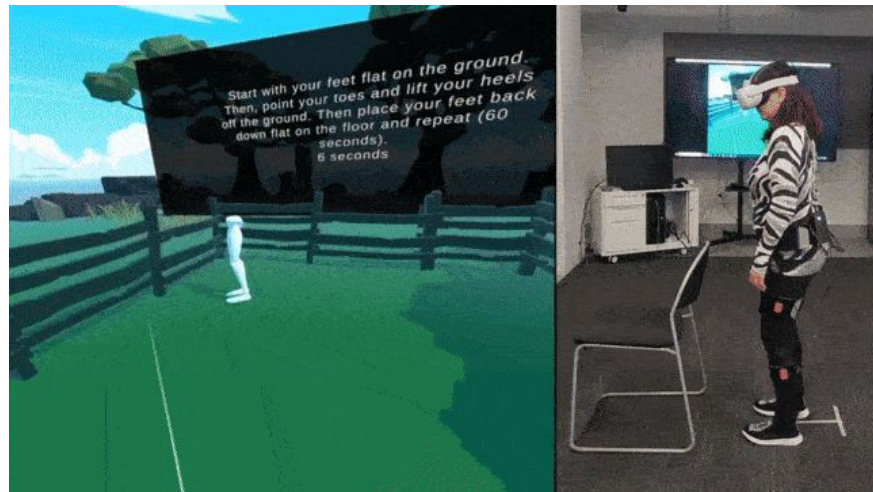
Limitations & Future Work

- No gamification or training progression –
 - ✓ Introduce *adaptive tasks*, *task difficulty scaling*, and *reward-based gamification* to sustain long-term motivation.
- Lower-body only tracking –
 - ✓ Integrate upper-body and trunk IMUs to enable *whole-body gait analysis*
- Clinical or mobility-impaired participants –
 - ✓ Run evaluations with: individuals with *gait disorders*, *patients in active rehabilitation*...to assess clinical relevance and therapeutic impact.



Summary

- **XRgait: a real-time gait training system combining wearable sensors + immersive displays**
 - Compared AR vs VR vs non-VR
 - Compared HeatMap vs SensorDot
 - Evaluated usability, engagement, and experienced.
- **Findings**
 - **VR > AR > non-VR** for engagement and presence
 - **HeatMap > SensorDot** for clarity
 - **Full lower-body visualisation** preferred
- **Implications**
 - Immersion significantly improves gait training experience
 - XR-based feedback can make rehab more engaging and measurable



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Acknowledgements :

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THANK YOU

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